

Ahmedabad University

Milestone 3(M3): Results and Analysis

Breast Cancer Survival Prediction

CSE516: Probabilistic Graphical Models

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1 Introduction and Problem Framing

1.1 Background and Motivation

Breast cancer is one of the most commonly diagnosed cancers worldwide, and understanding the factors associated with patient survival is an important aspect of cancer research. Advances in biomedical data collection have made it possible to study cancer using multiple sources of information, including genomic profiles and histological characteristics of tumour tissue. These sources provide different but complementary perspectives on the underlying biological characteristics of a tumour.

Genomic data captures molecular-level information, including patterns of gene expression and other molecular characteristics that may be associated with cancer development and progression. Histology-derived data, on the other hand, represents structural and morphological characteristics extracted from digitized tumour tissue images. While genomic information describes molecular patterns, histological features provide information about the organization and appearance of tumour tissue.

Although both modalities contain valuable information, analysing them independently may not capture the relationships that exist between molecular and morphological characteristics. Integrating these modalities can potentially provide a more comprehensive representation of the tumour and its associated survival patterns.

However, multimodal integration introduces several challenges. The two modalities differ in their feature representations and dimensionality, and identifying meaningful relationships between them is not straightforward. Furthermore, cancer survival data involves time-to-event information, where some patients may not have experienced the recorded event during the observation period. Therefore, an appropriate modelling framework must account for both the relationships between different data modalities and the nature of survival outcomes.

Probabilistic Graphical Models (PGMs) provide a framework for representing dependencies between variables through a structured probabilistic formulation. In particular, latent-variable models can represent shared information across multiple observed modalities using an underlying hidden representation. This provides a way to integrate genomic and histology-derived information while explicitly modelling uncertainty.

Motivated by these considerations, this project investigates a multimodal probabilistic framework for breast cancer survival prediction using genomic and histology-derived features from the TCGA-BRCA dataset.

1.2 Problem Statement

The central problem addressed in this project is the integration of genomic and histology-derived information for predicting breast cancer survival using a probabilistic graphical modelling approach.

Let:

- X represent the genomic feature matrix.
- Y represent the histology-derived feature matrix.
- Z represent a shared latent representation that captures information common to both modalities.

- T represent the observed survival time.
- δ represent the event indicator, distinguishing observed events from censored observations.

The project considers a latent-variable structure in which genomic and histology-derived observations are conditionally related through a shared latent variable:

$$X \leftarrow Z \rightarrow Y$$

The objective is to learn a probabilistic representation of the relationship between the two modalities and use the inferred latent representation for downstream survival prediction.

The problem can be divided into three interconnected tasks:

1. **Multimodal Representation Learning:** Identify shared information between genomic and histology-derived features through a latent-variable probabilistic model.
2. **Parameter Learning and Inference:** Estimate the parameters of the graphical model and infer the latent representation from the observed modalities using appropriate probabilistic methods.
3. **Survival Prediction:** Use the inferred latent representation as input to a survival modelling framework to estimate patient risk and evaluate survival prediction performance.

The project also investigates whether alternative parameter-learning procedures and different modality-availability conditions influence the quality of the learned representation and the resulting survival predictions.

1.3 Research Objectives

The primary objective of this project is to investigate the effectiveness of a multimodal Probabilistic Graphical Model for breast cancer survival prediction by integrating genomic and histology-derived information.

The specific objectives are:

1. **Reproduce the Existing Multimodal Framework:** Establish a working implementation of the reference probabilistic model using genomic and histology-derived features.
2. **Learn Shared Latent Representations:** Model the relationships between the two modalities through a common latent variable that captures shared information.
3. **Investigate Alternative Parameter-Learning Methods:** Compare the original CCA-based parameter estimation approach with an Expectation-Maximization (EM)-based learning procedure.
4. **Perform Latent-Variable Inference:** Investigate the inference of the shared latent representation using genomic-only, histology-only and combined multimodal observations.
5. **Predict Patient Survival:** Apply Cox Proportional Hazards modelling to the inferred latent representations and evaluate the resulting survival predictions.

6. **Compare Against Baseline Methods:** Evaluate the probabilistic approach against the implemented non-PGM baseline under consistent experimental conditions.

7. **Evaluate Robustness:** Investigate how the model behaves under changes in training sample size, input noise and modality availability.

8. **Analyse Experimental Performance:** Examine predictive performance, computational requirements, variability and limitations using appropriate quantitative and statistical evaluation methods.

1.4 Project Scope and Contributions

This project focuses on applying and evaluating an existing multimodal probabilistic modelling framework rather than proposing an entirely new graphical model.

The foundation of the implementation is a Gaussian Probabilistic Canonical Correlation Analysis (PCCA) model, which represents the relationship between genomic and histology-derived observations through a shared latent variable.

The project builds upon this framework through the following components:

1. **Multimodal Probabilistic Modelling:**

   The genomic and histology-derived feature matrices are jointly represented using a latent-variable model. This allows the framework to capture shared information between the two modalities rather than treating them as completely independent sources.

2. **Alternative Parameter Learning:**

   The original CCA-based parameter estimation approach is considered alongside an Expectation-Maximization-based learning procedure. The purpose is to investigate how the two approaches differ in parameter estimation, convergence behaviour and downstream predictive performance.

3. **Latent-Variable Inference:**

   The project investigates the estimation of the latent representation under different observation conditions:

$$P(Z \mid X, Y)$$

$$P(Z \mid X)$$

$$P(Z \mid Y)$$

   These conditional distributions represent inference using both modalities, genomics alone and histology alone, respectively. This comparison helps examine the contribution of each modality to the shared representation.

4. **Survival Modelling:**

   The inferred latent representation is used as input to a Cox Proportional Hazards model. This connects the multimodal probabilistic representation with a clinically relevant time-to-event prediction task.

5. **Experimental and Robustness Evaluation:**

   The implementation is evaluated against baseline methods and across different experimental configurations. Additional investigations consider missing-modality conditions, observation noise and variations in training sample size.

Together, these components provide an experimental framework for understanding the strengths, limitations and practical behaviour of probabilistic multimodal integration for breast cancer survival prediction.

1.5 Research Questions

The project is guided by the following research questions:

1. **RQ1:** Can genomic and histology-derived information be effectively integrated through a shared latent-variable probabilistic model for breast cancer survival prediction?

2. **RQ2:** How does EM-based parameter learning compare with the original CCA-based estimation approach in terms of convergence, computational behaviour and survival prediction performance?

3. **RQ3:** How does the availability of different modalities influence latent-variable inference and downstream survival prediction?

4. **RQ4:** Does combining genomic and histology-derived features provide an advantage over using either modality independently?

5. **RQ5:** How robust is the proposed framework to variations in training sample size, input noise and missing modality information?

6. **RQ6:** Under which experimental conditions does the probabilistic approach provide advantages over the implemented non-PGM baseline, and what limitations are revealed by the experimental evidence?

1.6 Focus of the M3 Evaluation

While the earlier milestone established the technical foundation and initial implementation of the project, the current milestone focuses on experimental evaluation and interpretation.

The M3 evaluation moves beyond demonstrating that the graphical model and prediction pipeline can be implemented. It examines how the framework performs under controlled experimental conditions, how the different learning approaches compare, and whether the observed results support the original research objectives.

Particular attention is given to:

- Comparing CCA-MLE and EM-based parameter learning.

- Evaluating multimodal, genomics-only and histology-only survival prediction.
- Measuring performance against the non-PGM baseline.
- Investigating variability across experimental folds and repeated runs.
- Examining robustness under different data and modality conditions.
- Analysing computational behaviour and model limitations.

The evaluation does not assume that the probabilistic model or the EM-based approach must outperform the baseline. Instead, the objective is to establish an evidence-based understanding of when multimodal probabilistic modelling is useful, where it performs less effectively, and what limitations remain.

The subsequent sections of this report describe the research foundation, dataset, probabilistic methodology, implementation and experimental design, followed by the results and their interpretation.

2 Research Background and SOTA Positioning

2.1 Research Background

The integration of multiple biomedical data modalities has become an important direction in computational cancer research. Different data sources describe different aspects of tumour characteristics, and combining them provides an opportunity to study relationships that may not be captured through individual modalities.

In this project, genomic and histology-derived features are considered as two complementary sources of information. Genomic features represent molecular characteristics, whereas histology-derived features describe morphological characteristics extracted from tumour tissue images.

A major challenge in multimodal analysis is identifying a common representation that captures the relationships between these different sources. Directly combining high-dimensional feature matrices may introduce redundancy and may not adequately represent the underlying dependencies between modalities.

Probabilistic latent-variable models provide a structured approach to this problem. Instead of modelling each modality independently, these models introduce an unobserved variable that explains the shared variation between the observed data sources.

The present project builds upon this idea through Gaussian Probabilistic Canonical Correlation Analysis (PCCA), which provides a probabilistic formulation for modelling the relationship between genomic and histology-derived observations.

2.2 Multimodal Probabilistic Modelling

Multimodal probabilistic modelling aims to represent the joint behaviour of multiple observed variables through a probabilistic framework.

Let X denote genomic observations and Y denote histology-derived observations. A shared latent variable Z is introduced to capture the common information present across both modalities.

The graphical structure is represented as:

$$X \leftarrow Z \rightarrow Y$$

Here, the latent variable Z acts as the common parent of the two observed modalities. Under this model, the modalities are conditionally independent given the latent representation:

$$P(X, Y \mid Z) = P(X \mid Z)P(Y \mid Z)$$

The corresponding joint distribution can be factorised as:

$$P(X, Y, Z) = P(Z)P(X \mid Z)P(Y \mid Z)$$

This formulation provides a probabilistic representation of the relationship between the two modalities.

Rather than directly combining genomic and histology-derived features, the model learns a shared latent representation that can be inferred from either modality individually or from both modalities jointly.

This property is particularly relevant to the project because it allows the contribution of each modality to be investigated through conditional inference.

2.3 Gaussian Probabilistic Canonical Correlation Analysis

Canonical Correlation Analysis (CCA) is a statistical method used to identify relationships between two sets of variables [**?**]. It finds linear combinations of the variables in each set that maximise their correlation.

Probabilistic Canonical Correlation Analysis extends this concept by introducing a latent-variable formulation. Both observed modalities are represented as linear transformations of a shared latent variable, together with modality-specific noise.

The model assumes a standard Gaussian prior over the latent variable:

$$Z \sim \mathcal{N}(0, I)$$

The two observation models are defined as:

$$X \mid Z \sim \mathcal{N}(W_x Z + \mu_x, \Psi_x)$$

$$Y \mid Z \sim \mathcal{N}(W_y Z + \mu_y, \Psi_y)$$

where:

- Z represents the shared latent variable.
- W_x represents the loading matrix associated with genomic observations.
- W_y represents the loading matrix associated with histology-derived observations.
- μ_x and μ_y represent the corresponding mean vectors.
- Ψ_x and Ψ_y represent modality-specific noise covariance matrices.

The latent representation provides a common space in which information from the two modalities can be analysed.

An important advantage of this formulation is that it allows posterior inference of the latent variable after observing one or both modalities.

For multimodal observations, the posterior distribution is:

$$P(Z \mid X, Y)$$

For observations from individual modalities, the corresponding posterior distributions are:

$$P(Z \mid X)$$

and

$$P(Z \mid Y)$$

These distributions provide the basis for comparing joint and modality-specific latent inference in our experiments.

2.4 Reference Framework and Existing Research

The project is based on the multimodal modelling framework investigated by Subramanian et al. (2024)[**?**]. The reference framework provides the foundation for integrating genomic and histology-derived information through correlated representations.

The original implementation considers multiple approaches to learning and representing relationships between the two modalities, including:

- Principal Component Analysis (PCA) for dimensionality reduction.
- Sparse Canonical Correlation Analysis (SCCA).
- Generalised Nonlinear Sparse Canonical Correlation Analysis (GN-SCCA).
- Different deflation strategies, including Hard Deflation (HD), Projection Deflation (PD) and Orthogonal Projection Deflation (OPD).

These approaches provide the comparative foundation for evaluating multimodal representations and their downstream survival prediction performance.

The reference implementation and its associated experimental results are used as the basis for reproducing the selected baseline configurations in our project.

Our work retains the original research framework as its foundation while investigating an alternative probabilistic parameter-learning procedure and additional inference and robustness experiments.

2.5 Reproduction of the Reference Framework

An important part of this project involves establishing a reproducible implementation of the existing framework before evaluating additional extensions.

The baseline implementation includes different feature representations and comparison methods. These include genomic-only, histology-only and concatenated multimodal representations.

The reproduced baseline also considers the original research configurations involving PCA, SCCA and GN-SCCA with their corresponding deflation strategies.

The baseline results provide a reference point against which the probabilistic model can be evaluated.

The reproduction focuses on maintaining consistent patient splits and evaluation conditions when comparing the different representations. The original authors' stored projection matrices are used for evaluating the reproduced paper-based representations; the original MATLAB training procedure itself is not claimed to have been independently rerun.

This distinction is important when interpreting the results because reproducing the evaluation using released model parameters is different from independently reproducing the complete original training process.

2.6 Positioning of the Current Implementation

The current project builds upon the existing multimodal probabilistic framework rather than introducing an entirely new graphical model.

The principal modelling structure remains:

$$X \leftarrow Z \rightarrow Y$$

The project investigates an alternative parameter-learning procedure through Expectation-Maximization (EM), alongside the existing CCA-based parameter estimation approach.

The comparison is intended to examine whether the alternative learning procedure can estimate the model parameters effectively and how differences in parameter estimation influence downstream survival prediction.

In addition, the project explicitly investigates latent-variable inference under different observation conditions.

The three inference settings are:

1. **Multimodal inference:** Both genomic and histology-derived observations are available.
2. **Genomics-only inference:** Only genomic observations are available.
3. **Histology-only inference:** Only histology-derived observations are available.

The inferred latent representations are subsequently used for survival prediction through Cox Proportional Hazards modelling.

The project also extends the experimental evaluation through ablation studies and robustness investigations. These include examining the effects of parameter configurations, training sample size, observation noise and modality availability, subject to experimental completion.

2.7 Revised SOTA Positioning

The project's position within the existing research landscape is based on the combination of an established multimodal probabilistic framework with additional parameter-learning, inference and experimental evaluation.

The project does not claim to introduce a fundamentally new probabilistic graphical model or to establish a new state-of-the-art result in breast cancer survival prediction.

Instead, its contribution is centred on the implementation and experimental investigation of alternative parameter learning and latent-variable inference within an existing multimodal PCCA framework.

The comparison between CCA-based estimation and EM-based learning provides an opportunity to study the behaviour of different parameter-learning approaches under the same underlying graphical model.

Furthermore, evaluating joint and modality-specific inference helps investigate the contribution of genomic and histology-derived information to the shared representation.

The final positioning of the project will be determined using the experimental evidence presented in the subsequent sections. In particular, the comparative performance, statistical variability, convergence behaviour and robustness results will be used to establish the actual advantages and limitations of the implemented approach.

Therefore, the project adopts an evidence-driven approach to evaluating multimodal probabilistic modelling rather than assuming that the proposed extension necessarily provides superior predictive performance.

3 Dataset and Preprocessing

3.1 Dataset Description

The project uses the publicly available The Cancer Genome Atlas Breast Invasive Carcinoma (TCGA-BRCA) dataset for investigating multimodal breast cancer survival prediction [**?**]. The dataset provides molecular and histological information associated with breast cancer patients, along with clinical information required for survival analysis.

The TCGA-BRCA dataset is suitable for this project because it provides multiple sources of information that can be used to study the relationship between tumour characteristics and patient survival.

Two primary data modalities are considered in the implementation:

1. Genomic features.

2. Histology-derived features.

These modalities are used as complementary observations within the probabilistic graphical model. Clinical survival information is used separately as the target information for downstream survival modelling.

The implementation works with processed feature representations rather than directly processing raw genomic sequences or histology images during model training.

3.1.1 Cohort Size and Patient Partitions

The processed cohort used in the project contains 974 patients. Of these, 95 patients experienced the recorded event, corresponding to approximately 9.8% of the cohort. The remaining observations are treated as censored records for survival analysis.

The dataset is organised into training, validation, and testing partitions as summarised in Table 1.

Table 1: Patient distribution across the dataset partitions.

Partition	**Patients**	**Percentage**
Training	585	60.06%
Validation	146	14.99%
Testing	243	24.95%
Total	974	100.00%

The training partition is used for model parameter estimation and downstream survival-model fitting. The validation partition is used for evaluation and configuration selection where applicable, while the testing partition is reserved for held-out performance evaluation. The five-fold experimental results reported later in this document should be interpreted according to the fold-generation and evaluation procedure used by the corresponding experiment scripts.

3.1.2 Feature Dimensions

Before dimensionality reduction, the genomic modality contains 1,000 features and the histology-derived modality contains 1,075 features. In the Paper PCA configuration, each modality is represented using 50 principal components.

Thus, the dimensionality-reduced matrices have the following dimensions:

$$X \in \mathbb{R}^{974\times 50}$$

$$Y \in \mathbb{R}^{974\times 50}$$

The concatenated representation contains 100 features per patient:

$$[X, Y] \in \mathbb{R}^{974\times 100}$$

The separate 50-dimensional modality representations are retained for joint and modality-specific inference, while the combined representation is used in the corresponding concatenation baseline.

3.2 Genomic Data

Genomic information provides molecular-level characteristics of the tumour. In this project, genomic data is represented through a numerical feature matrix containing molecular measurements associated with individual patients.

Let the genomic feature matrix be represented as:

$$X \in \mathbb{R}^{N\times D_x}$$

where:

- N represents the number of patients.
- D_x represents the number of genomic features.
- Each row corresponds to an individual patient.
- Each column represents a genomic feature.

The genomic matrix provides one of the two observed modalities used by the PCCA model.

These features are subsequently processed and represented in a form suitable for integration with the histology-derived modality.

3.3 Histology-Derived Data

Histological information describes the morphological characteristics of tumour tissue. In the original data collection process, histology information is obtained from digitized tissue images.

However, the model implemented in this project does not directly operate on raw histology images. Instead, it uses numerical feature representations derived from histological information.

Let the histology-derived feature matrix be represented as:

$$Y \in \mathbb{R}^{N\times D_y}$$

where:

- N represents the number of patients.

- D_y represents the number of histology-derived features.
- Each row corresponds to an individual patient.
- Each column represents a numerical feature extracted from histological information.

These features provide information about the morphological characteristics of tumour tissue and form the second observed modality in the graphical model.

The use of pre-extracted histology features allows the project to investigate multimodal integration without requiring the direct processing of high-resolution pathology images within the current implementation.

3.4 Clinical Survival Information

In addition to the two feature modalities, clinical survival information is required for the downstream prediction task.

The survival information consists of two important variables:

- **Survival Time (T):** Represents the observed time associated with each patient's survival record.
- **Event Indicator (δ):** Indicates whether the event of interest was observed or whether the patient's observation was censored.

These variables are used by the Cox Proportional Hazards model during survival analysis.

The survival information is not treated as a third observation modality in the PCCA structure. Instead, it is used for training and evaluating the downstream survival prediction model.

3.5 Multimodal Data Representation

The two feature modalities are represented as paired observations corresponding to the same patients.

The combined dataset can therefore be represented as:

$$\mathcal{D} = \{(X_i, Y_i, T_i, \delta_i)\}_{i=1}^{N}$$

where each patient has an associated genomic representation, histology-derived representation and survival record.

The paired representation is important because the probabilistic model learns the relationship between genomic and histology-derived observations belonging to the same patient.

The graphical model uses the two feature matrices as observed variables:

$$X \leftarrow Z \rightarrow Y$$

Here, Z represents the shared latent representation learned from the two modalities.

The resulting latent representation is subsequently used for downstream survival prediction.

3.6 Feature Configurations

Different feature configurations are considered to investigate how the representation of the input data influences model behaviour.

The project includes the reference-paper-based PCA representation and alternative feature configurations used during the experimental evaluation.

The principal configurations considered in the implementation include:

1. **Paper PCA Configuration:** Uses the feature representation associated with the reference-paper experimental setup.
2. **Alternative PCA Configuration:** Uses an alternative dimensionality-reduced representation for investigating the effect of feature configuration on model performance.
3. **Histogram-Based Configuration:** Uses histogram-derived feature representations where supported by the experimental pipeline.
4. **Log-Histogram Configuration:** Uses a logarithmically transformed histogram-based representation for investigating alternative feature distributions.

The configurations allow the model to be evaluated under different input representations while maintaining the overall multimodal modelling framework.

The exact feature dimensions and preprocessing details associated with each configuration are determined by the corresponding data-loading implementation.

3.7 Data Splitting Strategy

The dataset is evaluated using a five-fold experimental setup.

The data is organised into separate training, validation and testing partitions for each fold. This allows model parameters to be estimated using training observations, experimental configurations to be evaluated using validation observations where applicable, and predictive performance to be measured on held-out test observations.

For each fold, the data is represented as:

$$\mathcal{D}^{(k)} = \mathcal{D}^{(k)}_{\text{train}} \cup \mathcal{D}^{(k)}_{\text{valid}} \cup \mathcal{D}^{(k)}_{\text{test}}$$

where k denotes the fold index.

The training partition is used to estimate the PCCA parameters and fit the downstream Cox Proportional Hazards model.

The validation partition is available for model evaluation and configuration selection, depending on the experiment.

The testing partition is used to assess the generalisation performance of the trained model.

The use of consistent data partitions across compared methods is important because it ensures that observed performance differences are not simply caused by differences in the evaluation data.

The cohort-level partition counts reported earlier describe the processed dataset organisation. Fold-wise result tables in the experimental sections report the fold-specific metrics recorded by the corresponding experiment outputs; the fold indices should not be interpreted as additional patients or added to the cohort total.

3.8 Data Preprocessing and Standardisation

Preprocessing is an important step because genomic and histology-derived features may have different numerical scales and distributions.

To make the feature representations suitable for joint probabilistic modelling, the implementation applies standardisation using statistics estimated from the training partition.

For an individual feature j, the standardised value is calculated as:

$$x'_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}$$

where:

- x_{ij} represents the original feature value.
- μ_j represents the mean of feature j in the training data.
- σ_j represents the standard deviation of feature j in the training data.
- x'_{ij} represents the standardised feature value.

The same training-derived mean and standard deviation are applied to the corresponding validation and test features.

This prevents information from the validation and test partitions from influencing the preprocessing statistics used during model training.

The standardisation procedure is applied separately to the genomic and histology-derived feature matrices.

The resulting feature representations are then supplied to the PCCA model for parameter estimation and latent-variable inference.

3.9 Data Preparation for Survival Modelling

After preprocessing, the genomic and histology-derived observations are supplied to the probabilistic model.

The model estimates a shared latent representation using the available modalities.

For joint observations, the latent representation is inferred using:

$$Z_{\text{joint}} = \mathbb{E}[Z \mid X, Y]$$

For modality-specific observations, the corresponding representations are inferred using:

$$Z_{\text{genomics}} = \mathbb{E}[Z \mid X]$$

$$Z_{\text{histology}} = \mathbb{E}[Z \mid Y]$$

These inferred representations are used as input to the Cox Proportional Hazards model.

The corresponding survival time and event indicator are supplied separately during survival model training and evaluation.

This preparation establishes a common representation for evaluating survival prediction using multimodal and individual-modality observations.

3.10 Dataset Considerations and Limitations

Although the TCGA-BRCA dataset provides multiple sources of patient information, several considerations influence the experimental evaluation.

First, the genomic and histology-derived modalities have different feature characteristics and dimensionalities. Their integration therefore depends on the suitability of the learned shared representation.

Second, the dataset contains a finite number of patient observations. Consequently, the estimated model parameters and predictive performance may be sensitive to the available training sample size.

Third, survival analysis involves censored observations, which require appropriate treatment during model training and evaluation.

Finally, the use of pre-extracted histology-derived features means that the current implementation evaluates the integration of numerical representations rather than learning directly from raw histology images.

These considerations are taken into account during the experimental design and the interpretation of the results presented in the subsequent sections.

4 PGM Methodology

4.1 Overview of the Probabilistic Graphical Model

The proposed framework uses Probabilistic Canonical Correlation Analysis (PCCA) to model the relationship between genomic features and histology-derived features. Instead of treating the two modalities as independent sources of information, the model assumes that both are influenced by an underlying shared latent representation.

Let:

- X denote the genomic feature vector.
- Y denote the histology-derived feature vector.
- Z denote the shared latent variable.

The graphical structure of the model is:

$$X \leftarrow Z \rightarrow Y$$

Here, Z represents the unobserved shared factors that help explain the relationship between the two observed modalities. The model assumes that genomic and histology-derived observations are conditionally independent given the latent variable.

The corresponding joint probability distribution is factorised as:

$$P(X, Y, Z) = P(Z)P(X \mid Z)P(Y \mid Z)$$

This factorisation provides the probabilistic foundation for combining the two modalities through a common latent representation.

4.2 Latent Variable Representation

The latent variable Z captures the shared information between the genomic and histology-derived feature spaces. It is not a directly observed clinical variable or a ground-truth label. Instead, it is inferred from the available observations.

In the Gaussian PCCA formulation, the latent variable is modelled using a standard multivariate Gaussian prior:

$$Z \sim \mathcal{N}(0, I)$$

where I represents the identity covariance matrix.

The two observed modalities are modelled as linear transformations of the latent representation, together with modality-specific noise:

$$X \mid Z \sim \mathcal{N}(W_x Z + \mu_x, \Psi_x)$$

$$Y \mid Z \sim \mathcal{N}(W_y Z + \mu_y, \Psi_y)$$

Here, W_x and W_y are the loading matrices that connect the latent representation to the genomic and histology-derived feature spaces, respectively. The terms μ_x and μ_y represent

modality-specific means, while Ψ_x and Ψ_y represent the corresponding noise covariance matrices.

Through these relationships, the model learns a representation that captures information shared across both modalities while allowing each modality to retain its own characteristics.

4.3 Parameter Learning Using CCA-MLE

The model contains parameters that describe how the latent representation generates the observed genomic and histology-derived features. These include the loading matrices, modality-specific means, and noise covariance parameters.

The CCA-MLE approach estimates the model parameters using maximum likelihood estimation. The objective is to identify parameter values that make the observed multimodal data most likely under the assumed probabilistic model.

Conceptually, the estimation objective can be expressed as:

$$\hat{\theta}_{MLE} = \arg\max_{\theta} \log P(X, Y \mid \theta)$$

where θ represents the collection of model parameters.

The latent variable is not directly observed, so the likelihood involves accounting for the unobserved shared representation. CCA-MLE provides one parameter-estimation approach for the probabilistic canonical correlation framework.

4.4 Parameter Learning Using the Expectation-Maximization Algorithm

The Expectation-Maximization (EM) algorithm provides an alternative approach for estimating the parameters of the latent-variable model [**?**].

Since Z is unobserved, EM alternates between estimating the latent representation and updating the model parameters.

The algorithm consists of two main steps:

1. **Expectation Step (E-step):** Given the current parameter estimates, the model estimates the posterior distribution of the latent variable Z using the observed genomic and histology-derived features.

2. **Maximization Step (M-step):** Using the estimated latent-variable distribution, the model updates its parameters to improve the likelihood of the observed data.

These two steps are repeated until the algorithm satisfies its convergence criterion or reaches its maximum number of iterations.

The main distinction between the two learning approaches is therefore how the model parameters are estimated. Both approaches operate within the same underlying probabilistic framework, but use different parameter-estimation procedures.

4.5 Posterior Inference

After learning the model parameters, the framework can infer the latent representation for a patient using the available observations.

An important feature of the probabilistic formulation is that inference can be performed using both modalities or using only one modality.

4.5.1 Joint Inference

When both genomic and histology-derived features are available, the posterior distribution is:

$$P(Z \mid X, Y)$$

This posterior combines evidence from both modalities to estimate the shared latent representation.

Under the linear Gaussian formulation described above, the posterior covariance and mean can be expressed as:

$$\Sigma_{Z|X,Y} = \left(I + W_x^T \Psi_x^{-1} W_x + W_y^T \Psi_y^{-1} W_y\right)^{-1}$$

$$\mu_{Z|X,Y} = \Sigma_{Z|X,Y} \left[W_x^T \Psi_x^{-1} (X - \mu_x) + W_y^T \Psi_y^{-1} (Y - \mu_y)\right]$$

These expressions illustrate how information from both modalities contributes to the inferred latent representation under the stated Gaussian assumptions.

4.5.2 Genomics-Only Inference

When histology-derived information is unavailable, the model can infer the latent representation using genomic observations alone:

$$P(Z \mid X)$$

In this case, the posterior uses evidence from the genomic modality without requiring the histology-derived feature vector.

4.5.3 Histology-Only Inference

Similarly, when genomic information is unavailable, the model can use the histology-derived observations:

$$P(Z \mid Y)$$

This provides a way to obtain a latent representation even when only one modality is available.

The ability to perform joint and modality-specific inference is an important property of the probabilistic framework. It allows the model to be examined under different data-availability conditions.

4.6 Survival Prediction Using the Latent Representation

The inferred latent representation is used as an input to a downstream survival prediction model. In this project, the downstream model is the Cox Proportional Hazards model [**?**].

The Cox model represents the hazard function as:

$$h(t \mid Z) = h_0(t) \exp(\beta^T Z)$$

where:

- $h(t \mid Z)$ is the hazard at time t given the latent representation.
- $h_0(t)$ is the baseline hazard function.
- β is the vector of regression coefficients.
- Z is the inferred latent representation.

The model uses the latent features to estimate a patient's relative risk. The resulting risk score is based on the learned relationship between the latent representation and the observed survival outcomes.

The clinical survival information consists of the observed survival time and an event indicator. The event indicator distinguishes observed events from censored observations, where the exact event time is not known within the observation period.

The Cox model therefore connects the multimodal representation learned by PCCA to the project's final survival prediction task.

4.7 End-to-End Methodology

The overall methodology can be summarised as the following sequence:

1. Obtain the preprocessed genomic and histology-derived feature vectors for each patient.
2. Provide the two modalities to the PCCA framework.
3. Learn the shared latent-variable model using either CCA-MLE or the EM algorithm.
4. Infer the latent representation using joint or modality-specific posterior inference.
5. Use the inferred latent representation as input to the Cox Proportional Hazards model.
6. Evaluate the resulting survival risk predictions using the selected survival evaluation metrics.

This methodology establishes the connection between the probabilistic graphical model and the downstream survival prediction task. The graphical model provides a principled mechanism for representing shared information across genomic and histology-derived features, while the Cox model uses the resulting representation for survival-risk estimation.

5 Implementation and Experimental Design

5.1 Implementation Overview

The project implementation builds upon the multimodal probabilistic graphical model and baseline pipeline established during Milestone 2 (M2). The M3 stage focuses on evaluating the implemented methods, analysing their predictive performance, and investigating the behaviour of the model under different experimental conditions.

The implementation follows a modular workflow consisting of data preparation, model learning, latent-variable inference, survival prediction, and performance evaluation.

The main components of the pipeline are:

1. **Data preparation:** Load the preprocessed genomic, histology-derived, and clinical survival information.
2. **Model learning:** Estimate the parameters of the probabilistic model using the selected learning approaches.
3. **Latent-variable inference:** Infer the shared latent representation from the available modalities.
4. **Survival modelling:** Use the inferred representation for downstream survival-risk estimation.
5. **Evaluation:** Measure and compare the performance of the different configurations using the selected evaluation metrics.

The implementation retains the existing project pipeline as its foundation rather than introducing an unrelated modelling architecture.

5.2 Learning Methods Evaluated

The experimental evaluation considers three main approaches.

5.2.1 CCA-MLE-Based PGM

The CCA-MLE configuration represents the original parameter-learning approach used in the probabilistic multimodal framework. It serves as an important reference for evaluating the alternative learning procedure.

5.2.2 EM-Based PGM

The EM configuration applies the Expectation-Maximization algorithm to estimate the parameters of the latent-variable model. Its performance is compared against the CCA-MLE configuration to investigate the effect of using an iterative learning procedure.

The comparison considers not only survival-prediction performance but also the behaviour of the learning procedure during parameter estimation.

5.2.3 Non-PGM Baseline

A non-probabilistic baseline is included to examine whether the probabilistic modelling approach provides an advantage over a conventional feature-based approach.

This comparison is important because improvements over another PGM learning method alone would not establish whether the graphical modelling approach itself provides a practical benefit.

5.3 Experimental Evaluation Strategy

The experiments are organised to support comparisons between the learning approaches under consistent evaluation conditions.

A five-fold evaluation setup is used. The same fold structure is retained across the methods being compared so that differences in performance can be examined under corresponding data partitions.

For each fold, the methods are trained and evaluated using the corresponding experimental configuration. The resulting performance measurements are collected for fold-wise and aggregate analysis.

The evaluation reports the mean and standard deviation across the folds wherever applicable. This provides information about both the average performance and the variation observed across the evaluation partitions.

5.4 Baseline Comparison

A central part of the M3 evaluation is the comparison between the probabilistic methods and the non-PGM baseline.

The comparison is designed to answer the following questions:

- How does the CCA-MLE-based PGM perform relative to the non-PGM baseline?
- Does EM-based parameter learning produce comparable or different survival-prediction performance?
- Are the observed differences consistent across the evaluation folds?
- Does the choice of feature configuration influence the relative performance of the methods?

The methods are compared using the same evaluation metric and corresponding fold structure wherever the experimental setup permits. This helps reduce the possibility that performance differences arise simply from using different evaluation partitions.

5.5 Primary Evaluation Metric

The primary evaluation metric is Harrell's Concordance Index (C-index), which is commonly used to assess the ranking performance of survival prediction models.

The C-index measures how often the model assigns a higher predicted risk to the patient who experiences the event earlier among comparable patient pairs.

It can be represented conceptually as:

$$C = \frac{\text{Number of concordant comparable pairs} + \frac{1}{2}\text{Number of tied pairs}}{\text{Total number of comparable pairs}}$$

A higher C-index indicates better agreement between the predicted risk ordering and the observed survival outcomes.

The metric is particularly relevant to this project because the survival data include censored observations, where the exact event time is not known for every patient.

5.6 Robustness and Sensitivity Experiments

In addition to the primary baseline comparison, the M3 experimental plan investigates the robustness of the model under changes in data availability and input quality.

These experiments are intended to examine whether the model remains useful when the available patient information is incomplete or affected by noise.

5.6.1 Missing-Modality Evaluation

The missing-modality experiment investigates the effect of removing information from one of the two input modalities.

The planned conditions include:

- Complete genomic and histology-derived information.
- Missing histology-derived information.
- Missing genomic information.
- Increasing levels of missing modality information.

The objective is to examine how the availability of each modality affects latent-variable inference and downstream survival prediction.

5.6.2 Input Noise Evaluation

The noise-sensitivity experiment investigates how the model responds when perturbations are introduced into the input features.

Different noise levels are considered to examine whether the learned representation and survival-prediction performance remain stable as the input data become noisier.

5.6.3 Training Sample Size Evaluation

The sample-size experiment investigates the effect of reducing the amount of training data available to the model.

Different training-data fractions are considered while retaining the evaluation procedure defined for the experiment. The purpose is to understand whether model performance is sensitive to the amount of training information available for parameter estimation.

These robustness experiments are intended to provide a broader assessment than the primary comparison alone. Their results are reported separately from the main baseline evaluation.

5.7 Statistical Analysis and Result Aggregation

The experimental results are collected at the fold level and summarised to support comparisons across methods and configurations.

For each evaluated configuration, the analysis considers:

- Mean C-index across the evaluation folds.
- Standard deviation of the C-index.
- Fold-wise performance variation.
- Paired comparisons between methods evaluated on corresponding folds, where appropriate.

Statistical comparisons are interpreted cautiously because the number of evaluation folds is limited. In particular, repeated parameter initialisations are not treated as independent patient datasets.

The statistical analysis is therefore used to provide additional context for observed performance differences rather than to make stronger claims than the experimental design supports.

5.8 Experimental Reproducibility

To support reproducibility, the evaluation uses a consistent fold structure and records the results produced by individual experimental runs.

The analysis scripts are organised separately from the main model implementation. This separation allows the experimental outputs to be aggregated, compared, and visualised without modifying the underlying model code.

The resulting tables and plots provide the basis for the performance analysis presented in the following sections.

5.9 Summary

The M3 experimental design evaluates the existing multimodal PGM, compares alternative parameter-learning approaches, and investigates the effect of data availability, input noise, and training sample size.

The combination of baseline comparisons and robustness experiments is intended to provide a more complete understanding of the model's predictive performance, stability, and practical limitations.

6 Experimental Results and Baseline Comparison

6.1 Overview of Experimental Evaluation

This section presents the experimental results obtained from the multimodal Probabilistic Graphical Model (PGM) and compares its performance with the non-probabilistic baseline and reference baseline methods.

The evaluation focuses on survival-risk prediction using Harrell's Concordance Index (C-index). The experiments use the Paper PCA feature configuration and a five-fold evaluation setup. The compared approaches include CCA-MLE-based PGM, EM-based PGM, and a PCA-based feature-concatenation baseline.

The evaluation considers three input settings:

- **Multimodal:** Both genomic and histology-derived features are used for latent-variable inference.
- **Genomics-only:** Only genomic features are used for inference.
- **Histology-only:** Only histology-derived features are used for inference.

The reported C-index values are expressed as percentages. The mean and standard deviation are calculated across the five evaluation folds. Higher C-index values indicate better agreement between predicted risk rankings and observed survival outcomes.

6.2 Overall Multimodal Performance

The primary experiment compares the two PGM learning approaches against the PCA feature-concatenation baseline using both modalities.

Table 2 presents the overall performance.

Table 2: Multimodal survival prediction performance across five evaluation folds.

Method	**Mean C-index (%)**	**Standard Deviation**
CCA-MLE PGM	65.433	6.581
EM PGM	63.442	8.088
PCA Concatenation Baseline	66.082	6.966

The PCA-concatenation baseline achieved the highest mean C-index of 66.082%, followed by CCA-MLE at 65.433%. The EM-based PGM achieved a mean C-index of 63.442%.

The difference between CCA-MLE and the PCA baseline was approximately 0.649 percentage points, while the difference between EM and the PCA baseline was approximately 2.640 percentage points.

These results indicate that the multimodal PGM configurations did not outperform the PCA-concatenation baseline in the primary experiment. CCA-MLE achieved performance relatively close to the baseline, whereas EM produced a lower mean C-index.

The standard deviations also show variation in performance across the evaluation folds. EM exhibited the largest standard deviation among the three configurations, indicating greater fold-wise variability in this experiment.

6.2.1 Visual Comparison of Multimodal Performance

Figure 1 presents the multimodal C-index comparison across the evaluated methods.

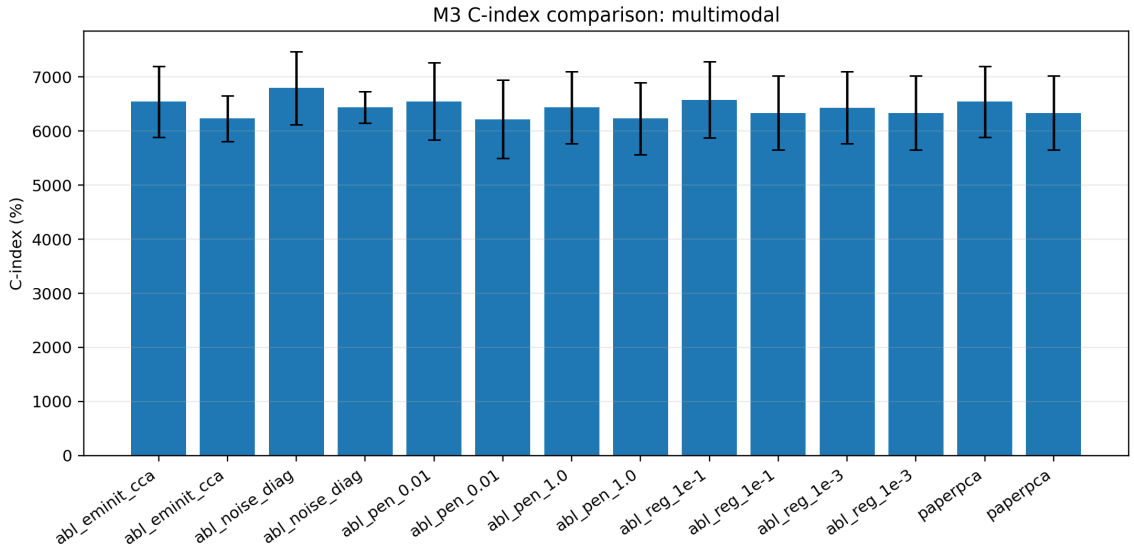


Figure 1: Multimodal C-index comparison across the evaluated methods.

The figure provides a visual comparison of the methods when both genomic and histology-derived information are considered. The observed results are consistent with the numerical comparison: the PCA-concatenation baseline achieves a higher mean C-index than both PGM configurations.

This finding suggests that the current multimodal latent-variable representation does not yet provide a measurable predictive advantage over the evaluated PCA-based feature-concatenation approach.

6.3 Fold-Wise Performance Analysis

To examine how the methods behave across individual evaluation partitions, the fold-wise multimodal C-index values are presented in Table 3.

Table 3: Fold-wise multimodal C-index values (%).

Fold	CCA-MLE	EM	PCA Baseline
0	61.045	67.704	75.980
1	69.393	67.463	65.809
2	61.401	52.529	59.689
3	76.478	73.982	71.485
4	58.850	55.531	57.448

The fold-wise results demonstrate that no single method achieves the highest C-index in every evaluation fold.

The PCA baseline performs particularly well in Fold 0, while CCA-MLE achieves the highest value in Fold 1 and Fold 3. The EM configuration achieves its highest value in Fold 3 but performs comparatively poorly in Fold 2 and Fold 4.

The variation across folds indicates that the relative performance of the methods depends on the evaluation partition. Therefore, the overall mean C-index should be interpreted together with the fold-wise results.

6.4 Performance Across Individual Modalities

In addition to multimodal inference, the model is evaluated using each modality separately. This experiment helps examine the individual contribution of genomic and histology-derived information to survival-risk prediction.

6.4.1 Genomics-Only Performance

Table 4 presents the results obtained using genomic features alone.

Table 4: Genomics-only survival prediction performance.

Method	Mean C-index (%)	Standard Deviation
CCA-MLE PGM	67.308	8.613
EM PGM	69.950	8.676
PCA Genomics Baseline	67.824	7.383

For genomics-only inference, EM achieved the highest mean C-index of 69.950%, followed by the PCA genomics baseline at 67.824%. CCA-MLE achieved a mean C-index of 67.308%.

These results indicate that genomic features provide useful information for survival-risk ranking under all three evaluated approaches. The EM configuration achieved the highest mean in this setting, although its performance variation across folds remains important when interpreting the result.

Figure 2 presents the visual comparison of the genomics-only results.

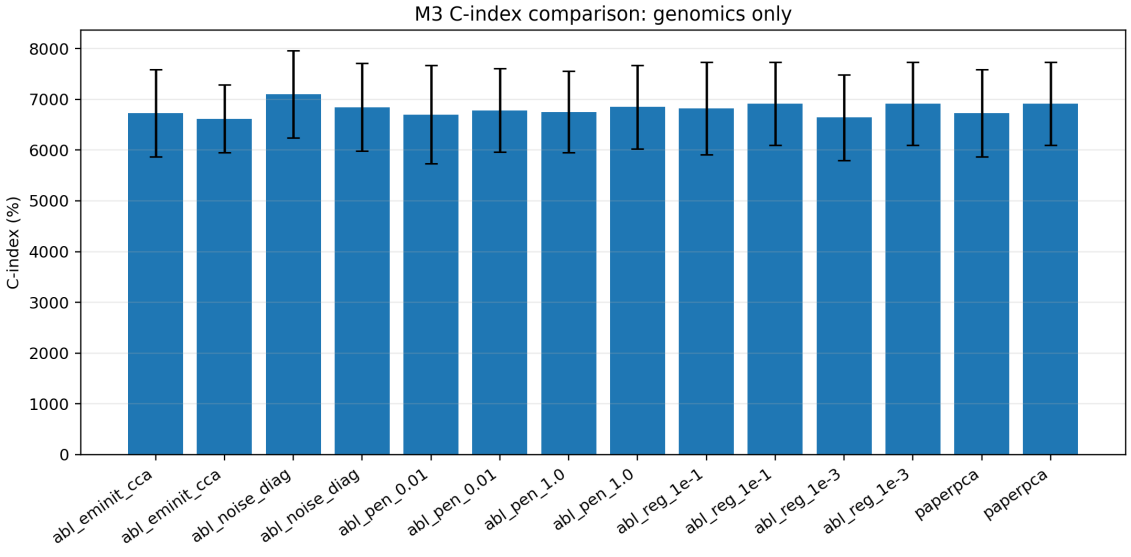


Figure 2: C-index comparison using genomic features only.

Table 5: Histology-only survival prediction performance.

Method	Mean C-index (%)	Standard Deviation
CCA-MLE PGM	57.339	6.455
EM PGM	56.928	8.380
PCA Histology Baseline	57.636	8.490

6.4.2 Histology-Only Performance

Table 5 presents the results obtained using histology-derived features alone.

The histology-only results are lower than the corresponding genomics-only results for all three methods.

The PCA histology baseline achieved a mean C-index of 57.636%, followed by CCA-MLE at 57.339% and EM at 56.928%.

The relatively lower values indicate that, under the current feature representation and experimental configuration, histology-derived features alone provide weaker survival-risk ranking performance than genomic features alone.

Figure 3 provides the corresponding visual comparison.

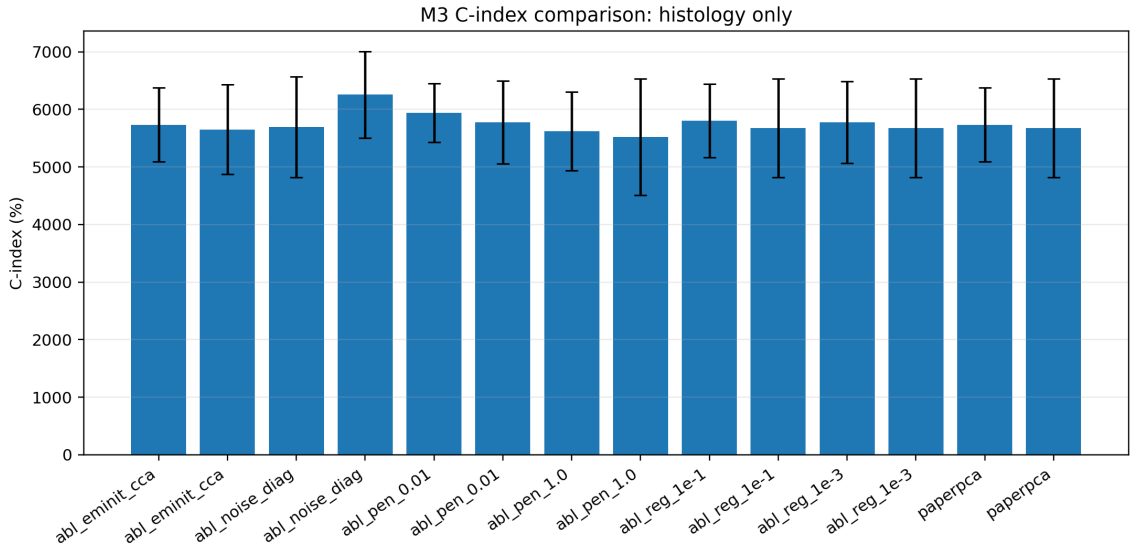


Figure 3: C-index comparison using histology-derived features only.

6.5 Comparison Between Modalities

The results across the three input settings reveal an important pattern.

For CCA-MLE, genomics-only inference achieved a mean C-index of 67.308%, compared with 65.433% for multimodal inference and 57.339% for histology-only inference.

For EM, genomics-only inference achieved 69.950%, compared with 63.442% for multimodal inference and 56.928% for histology-only inference.

The genomic modality therefore achieved the highest mean C-index for both PGM learning approaches.

An important observation is that combining genomic and histology-derived features did not improve the mean C-index over genomics-only inference in these experiments. This suggests

that multimodal integration does not automatically lead to improved survival prediction.

Although the PGM provides a shared latent representation of the two modalities, the contribution of histology-derived features may depend on the quality of their representation, the learning procedure, and the relationship between the two feature spaces.

The results therefore motivate further investigation into how the two modalities are integrated and how the shared latent representation is learned.

6.6 Comparison with Reference Baseline Methods

In addition to the primary PCA-concatenation baseline, the project includes a separate reference-baseline evaluation containing PCA, SCCA, and GN-SCCA configurations.

These methods are evaluated using genomic-only, imaging-only, and concatenated feature settings. The concatenated setting is particularly relevant because it provides a comparison with approaches that combine information from both modalities.

Table 6 presents the available concatenated-setting rerun results.

Table 6: Reference baseline methods in the concatenated feature setting.

Reference Method	**Mean C-index (%)**	**Standard Deviation**
PCA	66.47	6.86
SCCA/HD	67.79	8.26
SCCA/PD	67.58	4.69
SCCA/OPD	68.98	6.16
GN-SCCA/PD	65.47	10.14
GN-SCCA/OPD	66.85	9.47

Among the listed reference-baseline reruns, SCCA/OPD achieved the highest mean C-index of 68.98%, followed by SCCA/HD at 67.79% and SCCA/PD at 67.58%.

The PCA reference-baseline rerun achieved 66.47%.

These reference-baseline values are obtained from a separate baseline result file. They are therefore reported separately from the primary PCA-concatenation result of 66.082%, which belongs to the main Paper PCA experiment.

The two result sources should not be treated as identical runs. Their separate presentation preserves the distinction between the primary experimental comparison and the additional reference-baseline evaluation.

The available results show that several reference-framework methods achieved higher mean C-index values than the multimodal PGM configurations. In particular, the SCCA/OPD concatenated configuration achieved a higher mean C-index than both CCA-MLE and EM in the primary multimodal comparison.

For the GN-SCCA family, the available concatenated rerun results are reported for PD and OPD. The GN-SCCA/HD entries do not contain rerun results in the supplied baseline file and are therefore excluded from numerical comparison.

6.7 Summary of Main Experimental Findings

The experimental evaluation provides the following main observations:

1. **Multimodal performance:** The PCA-concatenation baseline achieved a higher mean C-index than both PGM learning approaches in the primary multimodal comparison.

2. **Genomics-only performance:** Genomics-only inference achieved the highest mean C-index for both CCA-MLE and EM. EM achieved the highest value among the three methods in this input setting.

3. **Histology-only performance:** Histology-only inference produced lower mean C-index values than genomics-only inference for all evaluated approaches.

4. **Fold-wise variation:** The ranking of the methods changed across evaluation folds, indicating that performance depends partly on the evaluation partition.

5. **Reference baseline comparison:** Several reference-framework methods, particularly SCCA/OPD in the concatenated setting, achieved higher mean C-index values than the PGM configurations.

6. **Multimodal integration:** The current experiments do not demonstrate that combining both modalities improves survival-risk ranking over using genomic features alone.

6.8 Discussion of Experimental Results

The results demonstrate that the proposed PGM can learn a shared representation from genomic and histology-derived features and use that representation for downstream survival-risk prediction.

However, the primary multimodal comparison does not show a predictive advantage over the PCA-concatenation baseline. The CCA-MLE configuration achieves a result close to the baseline, whereas EM achieves a lower mean C-index in the primary experiment.

The modality-specific evaluation provides further insight. Genomic features achieve stronger individual-modality performance than histology-derived features, and the highest mean C-index values are observed in the genomics-only setting.

These findings suggest that the current feature representations and integration strategy may not be extracting additional survival-relevant information from histology-derived features beyond what is already captured by the genomic modality.

The reference-baseline comparison also shows that some alternative feature-association methods achieve higher mean performance in the concatenated setting. This provides an important context for assessing the current PGM implementation and identifying areas for further improvement.

Overall, the experiments demonstrate the feasibility of the probabilistic multimodal modelling pipeline, but they do not establish that the current PGM configuration is superior to the evaluated non-PGM and reference baseline methods.

6.9 Limitations of the Experimental Results

The findings should be interpreted in light of the following limitations:

- The primary evaluation uses five folds, which provides a limited number of observations for estimating performance variability.

- The CCA-MLE and EM configurations use different latent dimensions in the saved experiment configuration: 30 for CCA-MLE and 50 for EM.

- The reference-baseline results are obtained from a separate result file and should not be directly treated as identical experimental runs to the primary comparison.
- The mean C-index alone does not establish statistical significance or generalisation to an independent patient cohort.
- The observed difference between genomic-only and multimodal performance requires further investigation before drawing conclusions about the biological contribution of either modality.

These limitations motivate the statistical, robustness, and ablation analyses presented in the following sections.

6.10 Section Summary

The primary multimodal experiment reports mean C-index values of 65.433% for CCA-MLE, 63.442% for EM, and 66.082% for the PCA-concatenation baseline.

The modality-specific experiments show that genomic features provide stronger individual-modality survival-risk ranking performance than histology-derived features under the evaluated configurations. The reference-baseline comparison further indicates that some alternative methods achieve higher mean performance than the current PGM configurations.

Taken together, these results establish the current predictive performance of the implemented multimodal framework and identify important areas for further investigation, particularly the contribution of histology-derived information and the effectiveness of multimodal integration.

7 Statistical Analysis and Uncertainty

7.1 Purpose of Statistical Analysis

The statistical analysis examines the consistency and variability of the survival-prediction results obtained from the evaluated methods.

Although the mean C-index provides an overall measure of predictive performance, it does not fully describe the variation across evaluation folds or establish whether the observed differences between methods are statistically meaningful.

Therefore, this section considers the following aspects:

- Mean and standard deviation of the C-index across evaluation folds.
- Fold-wise differences between the evaluated methods.
- Exploratory paired statistical comparisons.
- Sensitivity of EM performance to parameter initialisation.
- Limitations associated with the available evaluation design.

The analysis focuses primarily on the multimodal comparison between CCA-MLE, EM, and the PCA-concatenation baseline.

7.2 Descriptive Statistical Analysis

The mean and standard deviation of the C-index are used to summarise predictive performance across the five evaluation folds.

Table 7 presents the results from the primary multimodal experiment.

Table 7: Descriptive statistics of multimodal C-index across five evaluation folds.

Method	**Mean C-index (%)**	**Standard Deviation (%)**
CCA-MLE PGM	65.433	6.581
EM PGM	63.442	8.088
PCA Concatenation Baseline	66.082	6.966

The PCA-concatenation baseline achieved the highest mean C-index, followed by CCA-MLE and EM.

The standard deviation provides an indication of how much the C-index varied across the five folds. EM exhibited the largest standard deviation of 8.088 percentage points, suggesting greater fold-wise variability in the primary comparison.

CCA-MLE exhibited a standard deviation of 6.581 percentage points, while the PCA baseline had a standard deviation of 6.966 percentage points.

These values describe the observed variation across the evaluation folds. They should not be interpreted as confidence intervals or as direct estimates of uncertainty for performance on an independent patient population.

7.3 Exploratory Paired Statistical Comparisons

Since the evaluated methods use corresponding evaluation folds, paired comparisons can be used to examine the differences in their fold-wise C-index values.

An exploratory paired t-test was performed for the following comparisons:

1. CCA-MLE versus the PCA-concatenation baseline.
2. EM versus the PCA-concatenation baseline.
3. CCA-MLE versus EM.

For two methods, A and B, the difference in C-index for fold i is defined as:

$$d_i = C_{A,i} - C_{B,i}$$

The mean paired difference is:

$$\bar{d} = \frac{1}{n} \sum_{i=1}^{n} d_i$$

The paired t-statistic is calculated as:

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}$$

where s_d represents the sample standard deviation of the paired differences and n is the number of paired folds.

The comparisons use the five corresponding fold-wise observations from the primary experiment.

Table 8: Exploratory paired t-test comparisons using five evaluation folds.

Comparison	**Mean Difference (%)**	**95% CI**	t	p**-value**
CCA-MLE − PCA	-0.649	[-10.729, 9.431]	-0.179	0.867
EM − PCA	-2.640	[-8.772, 3.491]	-1.196	0.298
CCA-MLE − EM	1.992	[-4.927, 8.910]	0.799	0.469

The mean difference is expressed in percentage points. A negative value indicates that the first method achieved a lower mean C-index than the second method.

For the comparison between CCA-MLE and the PCA baseline, the mean paired difference is -0.649 percentage points. The corresponding p-value is 0.867.

For EM versus the PCA baseline, the mean paired difference is -2.640 percentage points, with a p-value of 0.298.

Finally, the comparison between CCA-MLE and EM produces a mean difference of 1.992 percentage points in favour of CCA-MLE, with a p-value of 0.469.

None of the three comparisons provides statistically significant evidence of a difference at the conventional 0.05 significance level.

The corresponding 95% confidence intervals also include zero, indicating that the observed fold-wise comparisons do not establish a consistent performance advantage for any of the methods.

7.4 Interpretation of Confidence Intervals

The confidence intervals in Table 8 describe uncertainty in the estimated mean paired differences between methods.

For example, the confidence interval for CCA-MLE versus the PCA baseline ranges from -10.729 to 9.431 percentage points.

This relatively wide interval indicates that the available five-fold evaluation provides limited precision when estimating the difference between the methods.

Similarly, the intervals for the EM comparison and the CCA-MLE versus EM comparison include both negative and positive values.

Consequently, although the observed mean differences favour the PCA baseline over both PGM configurations in the primary multimodal experiment, the statistical analysis does not establish that these differences are significant.

7.5 Sensitivity to EM Initialisation

The EM algorithm uses an iterative parameter-estimation procedure and may be influenced by its initial parameter values.

To investigate this behaviour, additional experiments were performed using five initialisation seeds: 42, 43, 44, 45, and 46.

Table 9 presents the multimodal C-index obtained for each seed.

Table 9: EM multimodal performance across different initialisation seeds.

Initialisation Seed	**Mean C-index (%)**	**Standard Deviation (%)**
42	63.442	8.088
43	63.576	8.049
44	64.111	6.309
45	61.144	5.651
46	64.402	6.101

The mean C-index varies across the five initialisation runs. The highest mean value is obtained with seed 46 at 64.402%, while seed 45 produces the lowest mean value of 61.144%.

The variation indicates that the predictive performance of EM is sensitive to the choice of initialisation.

This observation is relevant because the EM algorithm repeatedly updates its parameter estimates based on the current latent-variable distribution. Different starting points may therefore lead to different parameter-estimation trajectories and final representations.

However, these five runs use the same underlying evaluation structure. They represent repeated initialisation experiments rather than five independent patient datasets.

Accordingly, the variation across seeds is interpreted as evidence of initialisation sensitivity and not as an independent estimate of generalisation performance.

7.6 Limitations of the Statistical Evaluation

The statistical analysis has several limitations that affect the interpretation of the results.

1. **Limited number of evaluation folds:** The paired statistical comparisons use only five observations. This limits statistical power and makes the estimated differences sensitive to individual folds.

2. **Assumptions of the paired t-test:** The paired t-test assumes that the fold-wise differences are drawn from an approximately normal distribution. With only five observations, this assumption cannot be assessed reliably.

3. **Multiple comparisons:** Three exploratory method comparisons are reported. The resulting p-values are not adjusted for multiple comparisons and should not be treated as confirmatory evidence.

4. **Dependence between evaluation folds:** The fold-wise observations arise from the same overall dataset and evaluation procedure. They should not automatically be treated as fully independent samples from different populations.

5. **Repeated EM initialisations:** Different random seeds do not create independent datasets. They measure the sensitivity of the learning procedure to its initial parameter values.

6. **Absence of external validation:** The current statistical analysis does not establish generalisation to independent patient cohorts or external datasets.

For these reasons, the paired statistical comparisons are treated as exploratory analyses. No claim of statistical superiority is made based on the available results.

7.7 Overall Statistical Interpretation

The statistical analysis supports the descriptive findings from the primary experimental evaluation.

The PCA-concatenation baseline achieved the highest mean multimodal C-index, followed by CCA-MLE and EM. However, the exploratory paired comparisons did not identify statistically significant differences between the evaluated methods.

The confidence intervals are relatively wide, reflecting the limited precision of the five-fold comparison. In addition, the EM initialisation experiments demonstrate that the iterative learning procedure produces varying predictive performance under different starting conditions.

These findings highlight the importance of evaluating not only average predictive performance but also fold-wise variability and sensitivity to parameter initialisation.

Overall, the available evidence does not establish a statistically significant predictive advantage for the current probabilistic graphical model over the evaluated PCA-concatenation baseline. Additional independent validation and larger evaluation studies would be required to support stronger conclusions.

8 Robustness and Sensitivity Analysis

8.1 Objective of Robustness Evaluation

The robustness evaluation investigates how the proposed probabilistic graphical model behaves when the availability, quality, and quantity of input data are changed.

The primary objective is to understand whether the learned multimodal representation remains useful under different data conditions and to identify situations in which predictive performance deteriorates.

Three experiments are conducted:

1. **Missing-modality evaluation:** Examines the effect of randomly introducing missing information from one modality.
2. **Observation-noise evaluation:** Examines the effect of adding Gaussian noise to the input observations.
3. **Training sample-size evaluation:** Examines the effect of reducing the fraction of training data available for model estimation.

Both CCA-MLE and EM are evaluated using five folds. Harrell's Concordance Index (C-index) is used as the primary evaluation metric. The results are reported as mean C-index and standard deviation across the five folds.

8.2 Experimental Configuration

The robustness experiments use the Paper PCA feature configuration. The saved experiment configuration specifies the following settings.

Table 10: Configuration used for robustness experiments.

Parameter	**Value**
Feature configuration	Paper PCA
Evaluation folds	0, 1, 2, 3, 4
Learning methods	CCA-MLE and EM
CCA-MLE latent dimension	30
EM latent dimension	50
Random seed	42
Observation-noise model	Block
PGM regularisation	0.01
Cox penaliser	0.1
Maximum EM iterations	300
Missingness levels	0, 0.1, 0.25, 0.5
Noise levels	0, 0.1, 0.25, 0.5, 1.0
Training fractions	0.25, 0.5, 0.75, 1.0

An important configuration detail is that the saved experiment uses different latent dimensions for the two learning approaches: 30 for CCA-MLE and 50 for EM.

Therefore, the robustness results are useful for examining the behaviour of each method under changing experimental conditions, but direct differences between CCA-MLE and EM cannot be attributed exclusively to their learning algorithms.

8.3 Robustness to Missing-Modality Information

The first experiment investigates the effect of randomly introducing missing information from one of the two input modalities.

The experiment considers four missingness levels: 0, 0.1, 0.25, and 0.5. The zero-level condition represents the reference condition without additional missingness.

Table 11 presents the resulting performance.

Table 11: C-index under random one-modality missingness (%).

Method	Missingness	Mean	SD
CCA-MLE	0.00	65.34	7.39
CCA-MLE	0.10	65.07	7.37
CCA-MLE	0.25	65.99	6.25
CCA-MLE	0.50	60.33	8.93
EM	0.00	63.42	9.05
EM	0.10	63.00	8.76
EM	0.25	65.16	6.80
EM	0.50	61.08	8.32

For CCA-MLE, the mean C-index decreases slightly from 65.34% at the reference condition to 65.07% at a missingness level of 0.1. At level 0.25, the mean increases to 65.99%. However, at level 0.5, performance decreases to 60.33%.

A similar pattern is observed for EM. Its mean C-index changes from 63.42% at the reference condition to 63.00% at level 0.1. It increases to 65.16% at level 0.25 before decreasing to 61.08% at level 0.5.

The results indicate that both methods experience a noticeable reduction in performance at the highest tested missingness level.

The slight increases observed at level 0.25 should not be interpreted as evidence that missing information improves predictive performance. They may reflect variation across folds and the particular missingness realisations used in the experiment.

8.3.1 Probabilistic Inference When a Modality Is Unavailable

An important capability of the PCCA framework is that the shared latent representation can be inferred using the modality that remains available when the other modality is missing.

When both genomic and histology-derived features are available, joint inference uses:

$$P(Z \mid X, Y)$$

When histology-derived information is unavailable, the model can infer the latent representation from genomic observations:

$$P(Z \mid X)$$

When genomic information is unavailable, inference can instead use the histology-derived observations:

$$P(Z \mid Y)$$

In the missing-modality robustness experiment, the unavailable modality is masked for the affected observations, and inference is performed using the posterior associated with the available modality. The resulting latent representation is then passed to the downstream Cox Proportional Hazards model for survival-risk prediction.

This experiment therefore evaluates more than the effect of removing input features. It examines the use of modality-specific probabilistic inference to obtain a latent representation when complete multimodal observations are not available. This is a central capability of the PCCA formulation and an important component of the project's robustness evaluation.

8.3.2 Visual Analysis of Missing-Modality Robustness

Figure 4 presents the performance of the two learning approaches under different missingness levels.

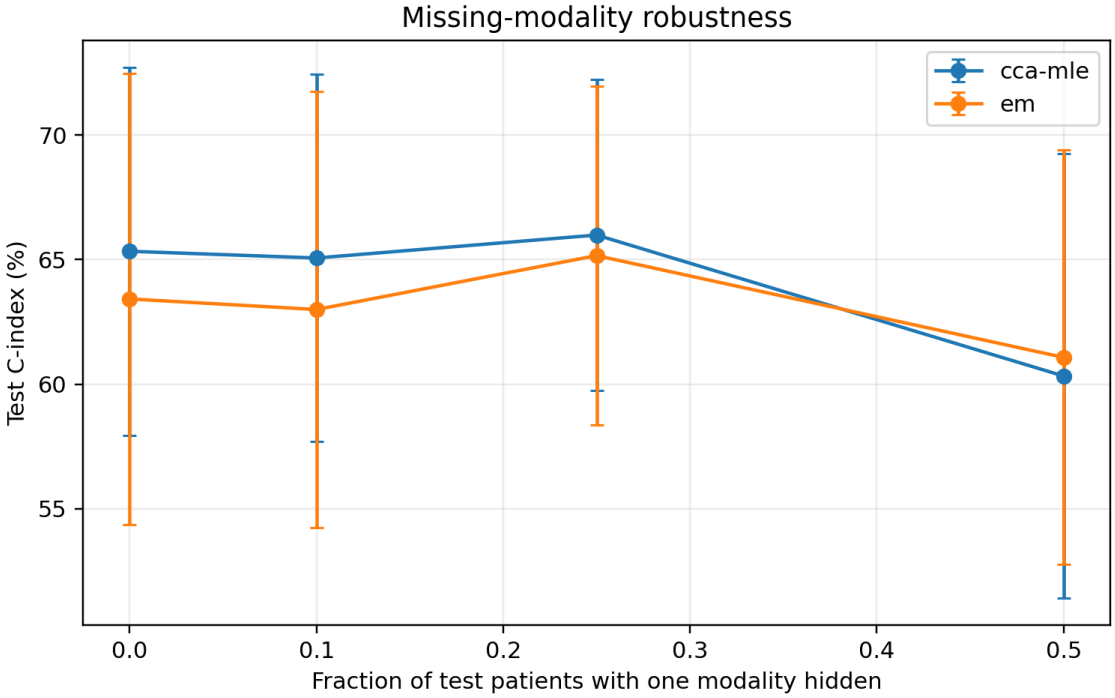


Figure 4: C-index performance under different levels of random one-modality missingness.

The figure illustrates that the two methods retain relatively similar performance at lower missingness levels, while both show a reduction at the highest tested level.

This suggests that the current framework can retain some predictive performance under partial modality unavailability, but its performance becomes weaker when the missingness level increases.

8.4 Robustness to Gaussian Observation Noise

The second experiment investigates how perturbations in the observed input features affect survival-prediction performance.

Gaussian observation noise is introduced at five levels: 0, 0.1, 0.25, 0.5, and 1.0. The zero-noise condition serves as the reference condition.

The results are presented in Table 12.

Table 12: C-index under Gaussian observation noise (%).

Method	Noise Level	Mean	SD
CCA-MLE	0.00	65.34	7.39
CCA-MLE	0.10	65.69	6.72
CCA-MLE	0.25	65.10	7.26
CCA-MLE	0.50	63.13	6.84
CCA-MLE	1.00	60.68	6.34
EM	0.00	63.42	9.05
EM	0.10	63.09	8.20
EM	0.25	62.35	11.13
EM	0.50	57.88	6.65
EM	1.00	53.21	2.92

For CCA-MLE, the mean C-index remains relatively close to its reference value at lower noise levels. It changes from 65.34% without additional noise to 65.69% at level 0.1 and 65.10% at level 0.25.

At higher noise levels, the performance decreases to 63.13% at level 0.5 and 60.68% at level 1.0.

The EM configuration shows a more pronounced decline as the noise level increases. Its mean C-index decreases from 63.42% at the reference condition to 57.88% at level 0.5 and 53.21% at level 1.0.

The total reduction between the reference condition and the highest noise level is approximately 4.66 percentage points for CCA-MLE and 10.22 percentage points for EM.

These results suggest that increasing observation noise negatively affects the predictive performance of both methods, with a larger decline observed for EM under the tested configuration.

8.4.1 Visual Analysis of Observation-Noise Robustness

Figure 5 presents the relationship between noise level and predictive performance.

The figure illustrates the reduction in predictive performance as the observation-noise level increases.

CCA-MLE maintains relatively higher performance under the tested noise conditions, whereas EM shows a more pronounced decline at higher noise levels.

This indicates that the learned representations are sensitive to perturbations in the observed features and that the degree of sensitivity differs between the two configurations.

8.5 Sensitivity to Training Sample Size

The third experiment investigates the effect of reducing the amount of training data available for parameter estimation.

Four training fractions are considered: 25%, 50%, 75%, and 100%. The results are evaluated across the five folds.

Table 13 presents the corresponding C-index values.

The results do not show a monotonic relationship between training-data fraction and predictive performance.

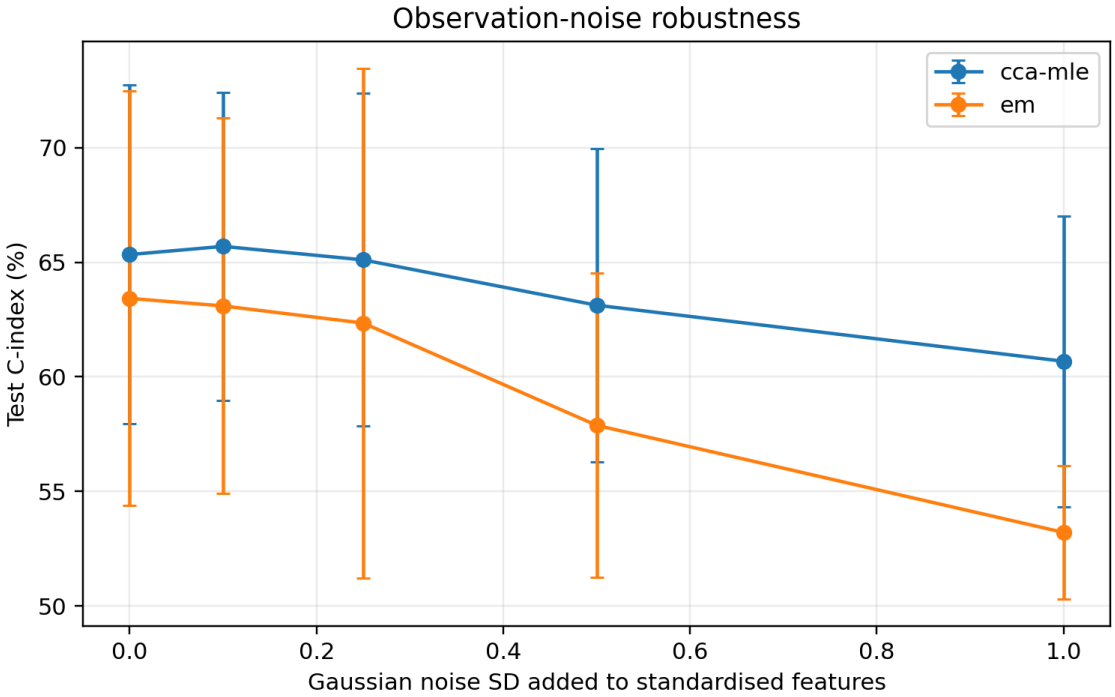


Figure 5: C-index performance under increasing levels of Gaussian observation noise.

Table 13: C-index under different training-data fractions (%).

Method	**Training Fraction**	**Mean**	**SD**
CCA-MLE	25%	67.16	6.36
CCA-MLE	50%	56.31	5.51
CCA-MLE	75%	63.46	5.41
CCA-MLE	100%	65.47	7.39
EM	25%	64.48	4.51
EM	50%	58.12	7.24
EM	75%	60.01	10.11
EM	100%	63.49	8.96

For CCA-MLE, the highest mean C-index is observed at the 25% training fraction, with a value of 67.16%. Performance decreases to 56.31% at 50%, increases to 63.46% at 75%, and reaches 65.47% when the full training fraction is used.

For EM, the highest mean C-index is also observed at the 25% training fraction, with a value of 64.48%. Its mean decreases to 58.12% at 50%, increases to 60.01% at 75%, and reaches 63.49% at the full training fraction.

Both methods achieve their lowest mean performance at the 50% training fraction.

The relatively high performance at the 25% fraction should not be interpreted as evidence that reducing the training dataset improves the model. The observed pattern may be influenced by the particular training subsets and evaluation folds used in the experiment.

The results instead indicate that predictive performance varies with the selected training fraction and that the relationship requires further investigation through repeated subsampling experiments.

8.5.1 Visual Analysis of Training-Data Sensitivity

Figure 6 presents the C-index across the different training-data fractions.

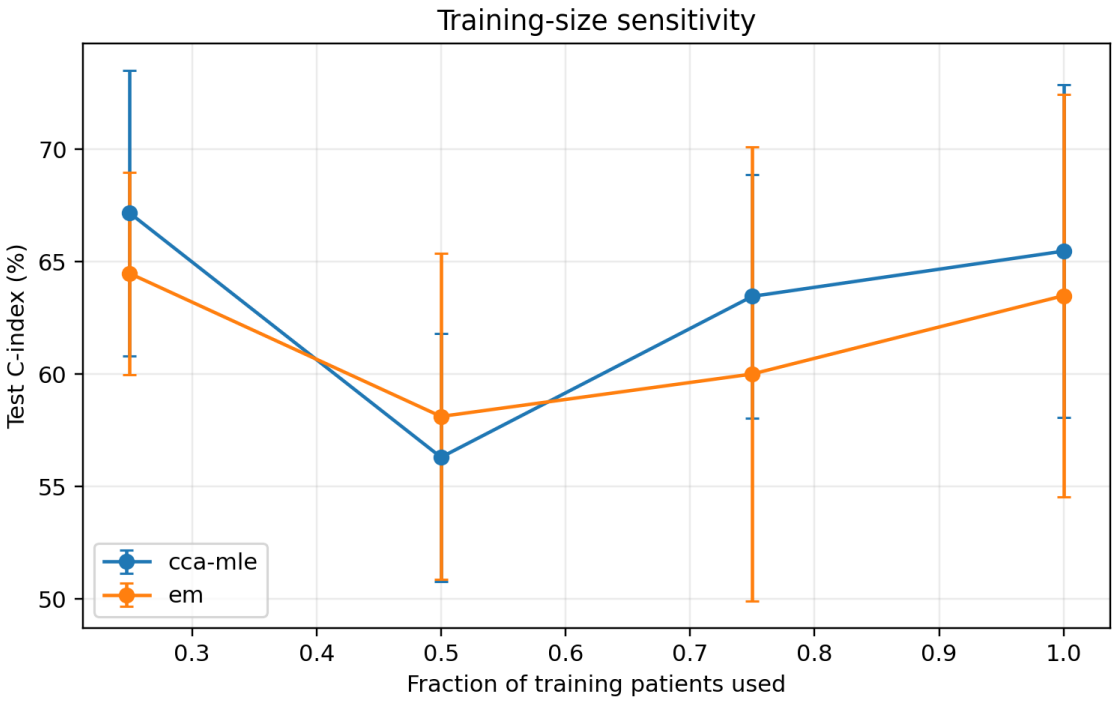


Figure 6: C-index performance under different training-data fractions.

The figure shows that the relationship between training-data availability and predictive performance is not consistently increasing.

Both learning approaches show a reduction in mean performance at the 50% training fraction, followed by an increase at higher fractions.

This non-monotonic behaviour suggests that the particular training subsets and fold composition may influence the results. It also indicates that additional repeated subsampling would be useful for understanding the effect of training sample size more reliably.

8.6 Comparative Interpretation of Robustness Experiments

The three experiments provide complementary information about the behaviour of the multimodal PGM.

The missing-modality experiment shows that both methods retain relatively similar performance at lower missingness levels, while their performance decreases at the highest tested level.

The observation-noise experiment shows a clearer decline as noise increases. The reduction is particularly noticeable for EM, whose mean C-index decreases to 53.21% at the highest noise level.

The training sample-size experiment shows non-monotonic performance changes. Both methods achieve their lowest mean C-index at the 50% training fraction, while their performance at the 25% fraction is higher than at the full training fraction.

Taken together, the results indicate that the current models are sensitive to changes in input availability, input quality, and training-data selection.

8.7 Limitations of Robustness Evaluation

Several limitations should be considered when interpreting the robustness results.

1. **Different latent dimensions:** The saved configuration uses a latent dimension of 30 for CCA-MLE and 50 for EM. Therefore, direct comparisons between the two methods are not fully controlled for latent dimensionality.

2. **Limited number of folds:** Each experimental condition is evaluated using five folds, which limits the precision of the estimated mean performance and standard deviation.

3. **Non-monotonic behaviour:** Some intermediate missingness and noise levels produce slightly higher mean C-index values than the reference condition. These observations should not be interpreted as evidence that the perturbations improve the model.

4. **Training subset variability:** The training-fraction results reflect the particular subsets used in the experiment. Repeated subsampling would be needed to assess whether the observed pattern is consistent.

5. **Perturbation assumptions:** The results are specific to the missingness and Gaussian-noise procedures implemented in the experiment. They do not establish robustness to every possible form of missing clinical or biological information.

6. **Limited scope:** These experiments do not establish robustness to external patient cohorts, distribution shifts, or alternative graphical structures.

8.8 Overall Findings

The robustness evaluation shows that the proposed multimodal PGM configurations retain predictive performance under some changes in data availability and quality, but their performance is affected by these changes.

Both methods show a reduction in mean C-index at the highest missingness level. Increasing Gaussian observation noise produces a clearer deterioration, particularly for EM. The training

sample-size experiment demonstrates non-monotonic performance, indicating that the amount of training data alone does not explain all observed variation.

The results also show that CCA-MLE maintains comparatively higher performance than EM under the tested observation-noise conditions. However, the different latent dimensions used in the saved configuration should be considered when interpreting this comparison.

Overall, the robustness experiments provide additional evidence about the sensitivity and limitations of the current implementation. They highlight the importance of considering data quality, modality availability, and training-data selection when evaluating multimodal probabilistic models for survival prediction.

9 Ablation and Error Analysis

9.1 Objective of Ablation Analysis

The ablation experiments investigate how different modelling and training choices influence the performance of the proposed probabilistic graphical model.

The main experiments evaluate the effect of changing:

- EM initialisation strategy.
- Observation-noise covariance structure.
- PGM regularisation strength.
- Cox Proportional Hazards regression penalisation.

These experiments provide additional insight into the sensitivity of the model to its configuration. In addition to predictive performance, the analysis considers EM iteration counts, likelihood behaviour, and other diagnostic information recorded in the experimental outputs.

The ablation experiments use the Paper PCA feature configuration and five evaluation folds. Five initialisation seeds, from 42 to 46, are available for each of the evaluated ablation configurations.

9.2 Effect of EM Initialisation Strategy

The first ablation investigates the effect of using CCA-based initialisation for the EM algorithm.

In this configuration, the EM algorithm is initialised using parameters obtained from the CCA-based approach. The purpose is to examine whether a more structured initialisation changes the final parameter estimates and downstream survival-prediction performance.

Table 14 presents the multimodal results.

Table 14: Effect of CCA-based initialisation on multimodal C-index.

Method	**Mean C-index (%)**	**SD (%)**
CCA-MLE PGM	65.433	6.581
EM with CCA initialisation	62.307	4.222
PCA Concatenation Baseline	66.082	6.966

The EM configuration with CCA-based initialisation achieved a mean multimodal C-index of 62.307%. This was lower than the CCA-MLE result of 65.433% and the PCA-concatenation baseline result of 66.082%.

The result indicates that using CCA-based initialisation did not produce an improvement in the downstream survival-prediction performance of EM in this experiment.

The standard deviation of 4.222 percentage points describes the variation across the five evaluation folds for this configuration. It is lower than the corresponding standard deviation of the primary seed-42 EM experiment, although the two configurations should be interpreted in the context of their respective initialisation procedures.

This finding suggests that initialisation strategy is an important experimental choice, but a structured starting point does not necessarily guarantee better predictive performance.

9.3 Effect of Observation-Noise Covariance Structure

The second ablation investigates the effect of changing the observation-noise covariance structure from block covariance to diagonal covariance.

The block-noise configuration is used as the reference, while the diagonal-noise configuration provides an alternative assumption about the structure of observation noise.

Table 15 presents the multimodal C-index results.

Table 15: Comparison of block and diagonal observation-noise configurations.

Method	**Block Noise (%)**	**Diagonal Noise (%)**
CCA-MLE PGM	65.433	67.973
EM PGM	63.335	64.388

The diagonal-noise configuration achieved higher mean C-index values than the block-noise reference for both learning approaches.

For CCA-MLE, the mean C-index increased from 65.433% to 67.973%. For EM, it increased from 63.335% to 64.388%.

These observations indicate that the assumed observation-noise covariance structure can influence the learned representation and its downstream predictive performance.

However, predictive performance alone is not sufficient to determine whether one covariance structure provides a better probabilistic model. The behaviour of the parameter-estimation procedure must also be considered.

9.3.1 EM Convergence Under Diagonal Noise

An important observation from this experiment is the number of iterations required by EM under diagonal noise.

The saved results report that EM reached the configured maximum of 300 iterations in every evaluation fold.

$$\text{Mean iterations} = 300$$

$$\text{Standard deviation} = 0$$

The likelihood traces in the saved results are recorded as monotonic, meaning that the reported likelihood values did not decrease during the iterations.

However, reaching the maximum iteration limit does not by itself establish whether the algorithm converged. It indicates that the procedure reached its configured computational limit in every fold.

The combination of the maximum iteration count and monotonic likelihood behaviour suggests that further investigation of the stopping criterion and likelihood progression is necessary.

Therefore, although diagonal noise produced higher mean C-index values in this experiment, the EM convergence behaviour remains an important limitation.

9.4 Effect of PGM Regularisation Strength

The third ablation investigates the effect of changing the regularisation parameter used in the PGM.

The reference configuration uses a regularisation value of 0.01. Two alternative values are evaluated:

$$\lambda = 0.001$$

and

$$\lambda = 0.1$$

Table 16 presents the corresponding multimodal results.

Table 16: Effect of PGM regularisation strength on multimodal C-index.

Method	**0.001**	**0.01**	**0.1**
CCA-MLE PGM	64.342	65.433	65.779
EM PGM	63.335	63.335	63.335

For CCA-MLE, the highest mean C-index among the three tested regularisation settings was obtained with $\lambda = 0.1$, at 65.779%. The lowest value was obtained with $\lambda = 0.001$, at 64.342%.

For EM, the mean C-index remained 63.335% at the displayed precision across the three tested regularisation settings.

These results indicate that changing the PGM regularisation strength produced different effects for the two learning approaches.

The CCA-MLE results show a small increase in mean C-index as the regularisation parameter increases across the tested values. In contrast, the displayed EM mean remained unchanged.

The findings do not establish a universally optimal regularisation value. They instead show that the effect of this parameter depends on the learning approach and should be considered alongside other model settings.

9.5 Effect of Cox Regression Penalisation

The fourth ablation investigates the effect of changing the penalisation strength in the downstream Cox Proportional Hazards model.

The experiment compares penaliser values of 0.01 and 1.0 against the reference value of 0.1.

Table 17 presents the multimodal results.

Table 17: Effect of Cox regression penalisation on multimodal C-index.

Method	**0.01**	**0.1**	**1.0**
CCA-MLE PGM	65.496	65.433	64.353
EM PGM	62.207	63.335	62.318
PCA Baseline	65.135	66.082	65.919

For CCA-MLE, the lower penaliser value of 0.01 produced the highest mean C-index among the three tested settings. Increasing the penaliser to 1.0 resulted in a lower mean value.

For EM, the reference penaliser value of 0.1 produced the highest mean C-index. Both the lower and higher penaliser configurations produced slightly lower values.

The PCA baseline also showed changes across the three settings. Its highest mean value was obtained with a penaliser value of 0.1.

These results indicate that downstream Cox regression penalisation influences the final survival-prediction performance. The setting associated with the highest mean C-index differs across the evaluated methods.

The results therefore highlight the importance of considering the downstream survival model when interpreting the performance of the complete multimodal pipeline.

9.6 Summary of Ablation Findings

The ablation experiments provide several observations about the current implementation.

1. **EM initialisation:** CCA-based initialisation produced a mean multimodal C-index of 62.307%. It did not outperform the CCA-MLE or PCA-concatenation reference results.

2. **Noise covariance:** The diagonal-noise configuration produced higher mean C-index values than the block-noise reference for both CCA-MLE and EM.

3. **EM iteration behaviour:** Under diagonal noise, EM reached the maximum configured iteration count of 300 in every fold. This raises a concern about the stopping behaviour and requires further investigation.

4. **PGM regularisation:** Changes in regularisation produced small differences in CCA-MLE performance, while the displayed aggregate EM mean remained unchanged across the tested values.

5. **Cox penalisation:** The downstream penaliser influenced predictive performance, and the highest-performing setting differed between CCA-MLE, EM, and the PCA baseline.

Overall, the ablation results demonstrate that the modelling assumptions and downstream survival-model settings can influence predictive performance.

9.7 Error Analysis and Model Diagnostics

The experimental result files contain several diagnostic measurements in addition to the C-index. These include training, validation, and test log-likelihood values, EM iteration counts, likelihood traces, fitting times, and posterior-entropy summaries.

These measurements provide additional information about the behaviour of the model during parameter estimation and inference.

In particular, the diagonal-noise experiment provides an important diagnostic observation because EM reaches the maximum configured iteration count in every fold. Although the recorded likelihood traces are monotonic, the iteration limit prevents us from concluding from the iteration count alone that the algorithm reached its convergence criterion.

The available diagnostic measurements therefore help identify potential issues in the parameter-estimation process, beyond what can be observed from the final C-index.

However, the supplied experimental outputs do not contain patient-level predicted risk scores, individual survival residuals, or detailed patient-level prediction errors.

Consequently, a detailed analysis of specific incorrectly ranked patient pairs or individual patient prediction failures cannot be established from the available result files.

The error analysis is therefore limited to the configuration-level and model-level diagnostics supported by the saved experimental outputs.

9.8 Overall Interpretation

The ablation analysis shows that the behaviour of the multimodal PGM depends on several modelling and training choices.

The diagonal-noise configuration produced higher mean predictive performance than the block-noise reference, but its EM implementation reached the maximum iteration limit in every fold. CCA-based initialisation did not improve the aggregated EM multimodal C-index. Changes in PGM regularisation and Cox regression penalisation also produced different effects across the evaluated configurations.

These findings demonstrate that the final survival-prediction performance depends not only on the choice of learning algorithm but also on the covariance assumptions, regularisation settings, initialisation procedure, and downstream survival-model configuration.

The results also highlight the importance of evaluating parameter-estimation behaviour alongside predictive performance.

Overall, the ablation experiments provide useful evidence about the sensitivity and limitations of the current implementation, particularly in relation to EM convergence and the selection of model hyperparameters.

10 Discussion and Revised SOTA Positioning

10.1 Overall Interpretation of Experimental Findings

The experiments conducted in this milestone provide a broader understanding of the behaviour of the proposed Gaussian Probabilistic CCA (PCCA) framework for multimodal breast cancer survival prediction. The results demonstrate that genomic and histology-derived feature representations can be integrated through a shared latent-variable model. However, the PGM-based approach does not consistently outperform the non-PGM PCA concatenation baseline.

In the M2 reference experiments, the PCA concatenation baseline achieved a multimodal C-index of $66.08 \pm 6.97\%$, compared with $65.43 \pm 6.58\%$ for CCA-MLE and $63.44 \pm 8.09\%$ for EM. These results indicate that the probabilistic latent-variable formulation does not automatically provide a predictive advantage over simpler feature concatenation.

The reduced-input experiments showed a different pattern in the comparison between the two PGM learning approaches. EM achieved a mean C-index of $67.58 \pm 7.76\%$, compared with $65.95 \pm 10.55\%$ for CCA-MLE. However, the non-PGM baseline achieved a higher mean C-index of $69.78 \pm 6.32\%$. Therefore, although EM performed better than CCA-MLE in this configuration, it did not outperform the non-PGM baseline. These results suggest that the relative performance of the learning approaches depends on the input configuration and that the probabilistic approach does not consistently provide an advantage over feature concatenation.

Overall, the findings highlight the importance of evaluating probabilistic modelling approaches against simpler alternatives under comparable experimental conditions.

10.2 Interpretation of the Latent-Variable Framework

An important contribution of the proposed framework is its representation of the relationship between genomic and histology-derived features through a shared latent variable Z.

Instead of directly concatenating the two modalities, PCCA models their relationship through a common latent representation. The latent variable is inferred from the observed modalities and represents information shared between them.

The experiments demonstrate that this modelling structure can be implemented and used for downstream survival prediction. However, predictive performance alone does not establish that the inferred latent representation captures biologically meaningful relationships or improves clinical interpretability.

Further investigation is required to evaluate the biological meaning, stability, and interpretability of the learned latent factors.

10.3 CCA-MLE and EM: Comparative Behaviour

CCA-MLE and EM provide two different estimation approaches within the probabilistic framework.

CCA-MLE provides a direct estimation approach, whereas EM iteratively estimates latent-variable-related quantities and updates model parameters. Their experimental behaviour suggests that the choice of estimation method affects predictive performance and computational behaviour.

Across the reference experiments, CCA-MLE generally provides competitive results and avoids some of the iterative stopping issues observed in EM configurations. EM can produce

competitive results in particular settings, but its performance and convergence behaviour are sensitive to the selected configuration.

The diagonal-noise ablation also showed that changing the noise covariance structure affects the results. In the tested configuration, diagonal noise produced higher average C-index values than block noise for both methods. However, this observation should be interpreted within the tested experimental setup rather than treated as a universal advantage of diagonal covariance.

10.4 Robustness and Sensitivity Findings

The robustness experiments indicate that model performance is affected by changes in input quality and training-data availability.

At higher Gaussian noise levels, EM showed a more noticeable decline in C-index than CCA-MLE in the tested configuration. At the highest noise level, the mean C-index decreased to 53.21% for EM and 60.68% for CCA-MLE.

The missing-modality experiment showed that both methods experienced a performance reduction at the highest tested missingness level. This suggests that the models can be affected when information from one modality becomes unavailable.

The training-fraction experiment also showed variation in predictive performance as the amount of training data changed. The results were not strictly monotonic, indicating that performance cannot be explained only by the proportion of available training data.

An important limitation is that the robustness experiments used different latent dimensions for CCA-MLE and EM. Consequently, their performance differences cannot be attributed exclusively to the estimation method.

10.5 Revised SOTA Positioning

The project initially aimed to reproduce and investigate a probabilistic multimodal framework for integrating genomic and histology-derived information. The M3 experiments extend this work through additional baseline comparisons, robustness experiments, and ablation studies.

Based on the available results, the proposed framework should be positioned as an experimental probabilistic modelling approach rather than as a demonstrated improvement over existing methods.

The current evidence supports the following conclusions:

- The Gaussian PCCA framework provides a structured approach for representing shared information across genomic and histology-derived modalities.
- Both CCA-MLE and EM can be integrated into the downstream survival-prediction pipeline.
- The non-PGM PCA concatenation baseline remains highly competitive.
- Robustness and ablation results show that modelling choices, input perturbations, and estimation procedures influence predictive performance.
- The current experiments do not establish statistically significant superiority of the proposed PGM over the non-PGM baseline.

Therefore, the contribution of this work lies primarily in implementing, evaluating, and analysing a probabilistic multimodal modelling pipeline, rather than claiming a new state-of-the-art survival-prediction result.

10.6 Limitations

The following limitations should be considered when interpreting the results:

1. **Predictive performance:** The PGM methods do not consistently outperform the non-PGM PCA concatenation baseline.
2. **Experimental variability:** The five-fold results show variability across folds, and the available experiments do not establish definitive statistical superiority.
3. **EM convergence:** Some EM configurations reached the maximum iteration limit, making convergence behaviour an important consideration.
4. **Configuration differences:** Certain comparisons use different latent dimensions or experimental settings, limiting direct attribution of performance differences.
5. **Patient-level analysis:** The available aggregated outputs do not provide sufficient patient-level predictions and residual information for detailed patient-level error analysis.
6. **Clinical interpretation:** The experiments evaluate predictive performance but do not establish clinical utility or biological validity of the learned latent representation.

10.7 Summary

The M3 evaluation provides a more complete assessment of the proposed probabilistic multimodal framework. It demonstrates the feasibility of the modelling pipeline and identifies the conditions under which its performance changes.

The results also reinforce an important experimental finding: a more complex probabilistic model is not necessarily a more accurate predictive model. The proposed PGM should therefore be viewed as a structured approach to multimodal representation learning whose predictive advantages require further investigation and validation.

11 Conclusion and Future Work

11.1 Conclusion

This project investigated the use of Probabilistic Graphical Models for multimodal breast cancer survival prediction by integrating genomic and histology-derived feature representations from the TCGA-BRCA dataset.

The proposed approach was based on Gaussian Probabilistic CCA, in which a shared latent variable represents the relationship between the two observed modalities. The latent representation was estimated using CCA-MLE and Expectation-Maximization (EM), and the resulting representations were used in a downstream Cox Proportional Hazards model for survival prediction.

The work progressed from establishing the technical foundation of the probabilistic model to evaluating its behaviour through baseline comparisons, robustness experiments, and ablation studies.

The experimental findings demonstrate that the proposed PGM pipeline can be implemented and evaluated for multimodal survival prediction. However, the results do not establish consistent predictive superiority over the non-PGM PCA concatenation baseline.

The robustness experiments further showed that predictive performance changes under input noise, modality missingness, and variations in training-data availability. The ablation experiments highlighted the influence of modelling and estimation choices on the resulting performance.

An important outcome of this work is therefore not only the implementation of a probabilistic multimodal approach, but also a systematic evaluation of its strengths, limitations, and sensitivity to experimental settings.

11.2 Key Contributions

The major contributions of the project are summarised below:

1. **Probabilistic multimodal modelling:** Implemented a Gaussian PCCA framework to represent shared latent information between genomic and histology-derived features.

2. **Parameter estimation:** Investigated CCA-MLE and EM as alternative estimation approaches within the probabilistic framework.

3. **Survival prediction pipeline:** Connected the inferred multimodal representations to a Cox Proportional Hazards model for predicting survival risk.

4. **Baseline evaluation:** Compared the probabilistic approaches with a non-PGM PCA concatenation baseline.

5. **Robustness evaluation:** Examined model behaviour under missing modality information, Gaussian observation noise, and reduced training-data availability.

6. **Ablation analysis:** Investigated the effects of selected modelling and regularisation choices on predictive performance.

11.3 Future Work

Several directions can be explored to improve the proposed framework and strengthen its evaluation.

11.3.1 Improved EM Convergence

Further work should investigate the convergence behaviour of EM, including alternative initialisation strategies, convergence thresholds, and maximum iteration settings. The likelihood progression and stopping conditions should be examined to distinguish genuine convergence from termination caused by reaching the iteration limit.

11.3.2 Consistent Hyperparameter Evaluation

Future experiments should compare CCA-MLE and EM under matched latent dimensions, regularisation settings, and data splits. This would make it easier to attribute performance differences to the estimation methods rather than to other configuration choices.

11.3.3 Advanced Missing-Modality Evaluation

The current implementation supports modality-specific probabilistic inference when one modality is unavailable. Further work could investigate more challenging missing-data conditions, including structured missingness patterns, cases where both modalities are partially unavailable, and situations where missingness is related to patient characteristics or data quality. Additional experiments could compare probabilistic inference with alternative missing-data handling strategies and evaluate performance across different missingness mechanisms.

11.3.4 Stronger Statistical Evaluation

Additional evaluation could include repeated cross-validation, confidence intervals, and appropriately paired statistical comparisons. These experiments would provide stronger evidence about the consistency of performance differences between the PGM and non-PGM approaches.

11.3.5 Latent Representation Analysis

The inferred latent factors could be examined to determine whether they capture meaningful shared patterns between genomic and histology-derived features. Such an investigation could include latent-factor stability, feature contributions, and associations with relevant biological characteristics.

11.3.6 Expanded Predictive Evaluation

Future work could include additional survival-prediction metrics and calibration analysis alongside the Harrell's C-index. Evaluation on an independent cohort would also be valuable for assessing the generalisability of the framework.

12 Repository Links

12.1 Project Repository

The source code, implementation, experimental outputs, and milestone-related materials are maintained in the project repository.

`https://github.com/shifaa174/CSE-516--PGM_Healthcare_brca`

12.2 Repository Organisation

The repository contains the project implementation and milestone-related folders.

- `M2_G02/`: Materials and results associated with the technical foundation milestone.
- `M3_G02/`: M3 experiments, results, and final submission materials.